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Toward sustainable urban design to promote walking activity: Nonlinear effects of multiple-scale urban design on pedestrian volume

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Abstract: Previous studies often explore the relationship between urban design and pedestrian volume based on linear assumptions, resulting in overestimation /underestimation. Using urban big data and air pollution data in New York City, this study employs explainable machine learning algorithms to uncover the nonlinear - interaction effects of urban design, air quality and vehicles on pedestrian volumes. Results show that macro-scale urban design plays a more important role than micro-scale urban design in determining pedestrian volume. Micro-scale urban design features contribute 14.26% to pedestrian estimation and impose nonlinear and threshold effects on walking volume. Traffic vehicle factors have nonlinear relationships with pedestrian volume, and the effective ranges and threshold identifications vary by variable. Ozone concentration leads to a reduction in walking volume if its concentration exceeds 27.3 ppb. This study provides valuable insights that enhance our understanding of the relationships between urban design, air quality, and pedestrian volume. It also offers guidance for urban planners on designing attractive neighbourhoods and implementing sustainable transportation planning.

Keywords: pedestrian volume; air pollution; traffic flow; multiple urban design; nonlinear relationship; machine learning, SHAP.

1. Introduction

Enhancing pedestrian volumes plays a pivotal role in fostering street vibrancy and promoting public health (Fuller and Moore, 2017). Walking, a ubiquitous form of physical activity, benefits active lifestyles and offers social advantages such as reducing private vehicular use, mitigating pollution, and facilitating social engagement (Foster et al., 2018; Lee et al., 2023). Furthermore, consistent engagement in walking contributes to improved human health, including mitigating obesity risk, cardiovascular enhancement, and heightened cognitive function (Vogel et al., 2009).

Promoting walking and increasing pedestrian volume has become a primary focus in public health and urban planning (Chen et al., 2020a; He et al., 2021). Beyond individual-focused educational interventions, there is growing attention to the role of the built environment (Forsyth et al., 2007; Gao et al., 2019), in encouraging walking activities. Numerous studies have substantiated the association between the built environment and walking activities (Lu et al., 2018; Yang et al., 2021a; Zhou et al., 2024). For example, Li et al. (2005) identified a positive relationship between the built environment and walking activity at the neighbourhood scale. Handy et al. (2002) emphasized combining urban design, land use, and transportation systems to promote walking activities. Despite these advancements, many studies rely on individual walking behaviours assessed through surveys and questionnaires, which entail limitations such as time consumption and potential social desirability bias. Moreover, research efforts have often focused on the neighbourhood and street segment scales (Chen et al., 2022a). Qualitative studies have also provided insights into how micro-scale urban design and air pollution can influence physical activities like walking and cycling. A well-designed streetscape attracts both locals and visitors to engage in walking activities, while increased air pollution concentration discourages outdoor activities. Despite the influence of various factors on walking has been widely acknowledged, the question of how micro-scale urban design, vehicle flow, and air quality, impact pedestrian volume remains a subject of considerable attention.

Scholars have delved into the relationship between urban design and walking activities (Cao and Tao, 2023; Li and Ma, 2022; Wu et al., 2023). For instance, Liu et al. (2023a) explored the association between street view and walking duration in Amsterdam, the Netherlands, suggesting that walking policies and urban planning need to consider specific street environments. Similarly, Ki and Lee (2021) assessed how the green view index of neighbourhood streets affects walking time, indicating that the green view index is more closely associated with walking time than traditional greenery variables. However, these studies have examined factors (socio-economic, transportation, urban streetscape) influencing pedestrian volumes based on linear or log-linear assumptions (Barnett et al., 2017; Chen et al., 2022a; Gao et al., 2020; Guzman et al., 2020). Depending on linear algorithms may lead to overestimating the impact of explainable variables (Yang et al., 2021a).

Recently, numerous studies have employed nonlinear algorithms such as machine learning to unveil the intricate relationships in urban analytics (Cheng et al., 2020; Doan et al., 2024; Liu et al., 2021; Xiao et al., 2021). Scholars have utilized gradient-boosting decision trees (Liu et al., 2021), random forest (Yang et al., 2021a), extreme gradient boosting (Ji et al., 2022; Wang et al., 2021) to explore nonlinear threshold effects in the field of urban analysis. Unlike traditional models such as OLS and GWR, machine learning exhibits greater flexibility and performance estimation (Doan et al., 2024; Liu et al., 2021). However, machine learning is often considered a black box, making it challenging to understand why specific results emerge, such as threshold effects and interactions.

Despite these efforts, there are still research gaps that require further investigation. Firstly, most studies focus on single-scale urban design (Cheng et al., 2020; Cheng et al., 2022; Wu et al., 2021), lacking consideration of multiple urban design factors and their impact on pedestrian volume. Single-scale urban design may not comprehensively explain the relationship between urban design and pedestrian volume, potentially resulting in ineffective recommendations for urban planners. Secondly, although air pollution and traffic flow can individually affect walking activities (Giallouros et al., 2020), the mechanistic effects of air pollution, vehicle flow and urban design on pedestrian volume are rarely considered simultaneously. Neglecting these crucial features may lead to biased results since it cannot account for the interaction effects. Thirdly, the nonlinear association between traffic volumes and air quality concerning pedestrian volume remains unknown. The critical issue is to what extent an increase in air pollution concentration and traffic flow leads to a decrease in walking volume and what the threshold effects are between pedestrian volume and air quality. Notably, increased pollution does not immediately result in decreased pedestrian volume. Beyond a certain threshold, changes in pollution concentration result in pedestrian volume fluctuations. Linear assumptions may lead to overestimating the impact of explanatory variables (Yang et al., 2021a). Therefore, further investigation is warranted into the nonlinear relationship, threshold effects, and local and global interactions (Cheng et al., 2020; Liu et al., 2021; Yang et al., 2022). To the best of our knowledge, this is the first study to examine the nonlinear relationships and interaction effects between multiple-scale urban design features, traffic flow, air pollution, and pedestrian volume.

To address the aforementioned knowledge gaps, we investigated the threshold effects of multiple-scale urban design, vehicles and air pollution on pedestrian volume utilizing various machine learning algorithms and SHAP (Shapley Additive exPlanations). Specifically, we employed deep learning techniques to extract pedestrian volume from over 29,540 street view images, which were then aggregated with the multiple urban design and air pollution data. Subsequently, we applied machine learning algorithms and SHAP as an explainable technique to examine threshold effects, nonlinear mechanisms, relative importance, and interactions among explanatory variables influencing pedestrian volume. The key contributions of this study are as follows:

(1) Identifying the relative importance of urban design, air pollution and vehicles in influencing pedestrian volume.

(2) Uncovering the nonlinear mechanisms of multiple urban design characteristics, traffic flow features, air pollution, and vehicle flow on pedestrian volume, elucidating global to local interactions using urban big data and machine learning.

(3) Providing recommendations for urban designers and policymakers to foster walking activities and contribute to sustainable urban design.

The remainder of this work is structured as follows. Section 2 provides a literature review of the relationship between urban design and pedestrian volume, from linear to nonlinear. Section 3 presents a conceptual framework. Section 4 describes the study area, study design, data collection and methodology. Section 5 presents the main results, including the relative importance of explanatory variables and nonlinear-interaction effects. Section 6 discusses the results and their implications in urban planning and transportation, followed by the conclusion.

2. Related works

Previously, numerous studies have explored the relationship between urban design and pedestrian volume (Guzman et al., 2020; Yang et al., 2021a; Zhao and Wan, 2020). These studies can be categorized into two groups: linear approaches (Ahmadipour et al., 2021; Zhao and Wan, 2020) and nonlinear approaches (Duan et al., 2023; Guo et al., 2023). Most existing studies have employed linear assumptions, such as OLS, structural equation models (Yang et al., 2023; Zafri and Khan, 2022), and logistic regression Porter et al. (2018), to identify the relationship between built environment features and walking activities (Chen et al., 2020b; Khan, 2023; Soto et al., 2023). For example, using logistic regression, Malambo et al. (2017) explored the association between built environment features and walking time. The results showed significant effects of the built environment on walking activities. Additionally, prior studies have utilized spatial regression models, such as Durbin models (Jiang et al., 2021), to explore the spatial dependence and heterogeneity of urban design features on walking activities. Several studies have explored the geographical effects of urban design on pedestrian volume using GWR (Yin and Zhang, 2021; Zafri and Khan, 2022). For example, Kim et al. (2019) found a spatially varying relationship between the built environment and walking using the GWR model. However, the findings of these studies mostly demonstrate linear relationships. In reality, nonlinear relationships between urban variables and pedestrian volume always exist, as an increase in explanatory variables may not lead to a linear increase or decrease in the target variable. Contrary to linear assumptions, escalating determinants would not necessarily lead to proportionately inflated target variables, potentially resulting in implausibly high estimations (Chen et al., 2022b; Doan et al., 2024; Liu et al., 2021).

Recently, researchers have shifted their focus towards exploring complex nonlinear relationships, utilizing nonlinear assumptions to overcome the limitations of linear models. For instance, scholars have employed machine learning algorithms such as GBDT, RF, and XGBoost to unveil the nonlinear impacts of the urban built environment on physical activities such as walking and cycling (Guo et al., 2023; Liu et al., 2023b; Yang et al., 2024b). These studies have affirmed the presence of nonlinear relationships and have identified thresholds in built environment variables. For example, Liu et al. (2021) utilized GBDT to investigate the nonlinear association between the built environment and working and shopping activities, revealing that the built environment holds greater importance than socio-economic factors in influencing active travel. In another study, Cheng et al. (2020)

explored the nonlinear effects of the built environment on elderly individuals' walking behaviours, highlighting threshold effects in transportation characteristics. However, most of these studies focus on the effects of macro-scale urban design on walking activities, often overlooking micro-scale urban design factors. To address this limitation, recent works have utilized open-sourced street view images (such as Google Street View and Baidu Street View) to assess streetscape design and its association with walking behaviour (Yang et al., 2024a; Yang et al., 2021a). For example, Liu et al. (2023b) highlighted that openness contributes between 7.6% and 8.4% to jogging behaviours. In another study, Yang et al. (2021b) investigated the impact of street greenery on the walking time of the elderly, revealing a significant influence in suburban areas. However, these prior studies often do not integrate crucial factors affecting pedestrian volumes, such as pollution and road vehices and multiple urban design features. Research has indicated that air pollution may diminish the benefits of active travel (Tainio et al., 2016) and decrease pedestrian volume (Chung et al., 2019). Thus, omitting these features may lead to an incomplete understanding of the interaction effects between explanatory and pedestrian volume.

3. Conceptual framework

The conceptual framework for empirical analysis is illustrated in Figure 1, which unpacks the relationship between multi-scale urban design, air quality, road vehicles, and pedestrian volume. Multi-scale urban design encompasses both micro-level and macro-level urban design. Macro-level urban design addresses fundamental activity needs, providing spaces and pathways for human activities. Consistent with previous studies, macro-scale urban design is measured through a built environment framework. The micro-level urban design focuses on urban design quality, considering factors such as visual greenery, pavement ratio, sky ratio, and building ratio.

Research has highlighted the impact of air quality on people's awareness and behaviour (Marshall Julian et al., 2009). When air quality is low, pedestrians may enjoy walking and shopping on the street, but sensitivity increases as air quality surpasses standard levels (Doan et al., 2025). Therefore, air quality is included to quantify the interaction effects between urban design and air pollution on pedestrian volume. Additionally, acknowledging the significant role of various road vehicles (such as buses, motorcycles, and bikes) in promoting pedestrian flow (Rissel et al., 2012), this study examines how different road vehicles interact with urban design and quality to influence pedestrian volume.

This work explicitly explores the pathways through which urban design, pollution, and traffic vehicles influence pedestrian volume. While previous studies have separately examined the significant contributions of both micro and macro-level urban design, this research delves into potential nonlinear-interaction effects among urban design features, road vehicles, and air quality. These findings contribute to a deeper understanding of walking behaviour promotion, offering new insights for sustainable urban design to enhance walkable spaces.

Figure 1. Conceptual framework

4. Materials and methods

4.1. Study area

We chose the Manhattan district, the core area of New York City (NYC), as the focal point of our case study. Manhattan, renowned as NYC's commercial hub, covers an area of 87 km^2 and had a population of 1.69 million in 2020. In this district, a significant portion of urban built-up areas, population centres, and socio-economic activities are concentrated. Manhattan boasts well-established leisure facilities, commercial hubs, and green spaces, making it an ideal walking setting. Moreover, the district features attractive urban design and convenient public transportation, fostering a conducive environment for pedestrian engagement.

4.2. Study design

Figure 2 depicts the analytical flow of the study, comprising three key steps: (1) data collection and processing; (2) variable selection and modelling of nonlinear associations; (3) interpretation of nonlinear threshold effects. Initially, pedestrian volume extracted from Street View Images (SVI) is the dependent variable. Traffic vehicles—such as buses, cars, motorbikes, and bikes—counted from SVI as explanatory variables. The micro-urban design, encompassing sky, road, trees, buildings, and pavement ratio, is extracted using semantic segmentation. Air pollution data, including ozone, NO, and $PM_{2.5}$, is collected from NYC open data. Additionally, geospatial data is employed (from MapPLUTO and NYC open data) as macro-level urban design, encompassing building characteristics, urban amenities, accessibility, and location characteristics, as these elements have demonstrated close associations with pedestrian volume. As in previous studies, all variables are integrated into a grid of 100x100m (Huber, 2021; Salazar-Miranda et al., 2023). After data cleaning, 5273 grids remained for analysis.

In the second step, an Ordinary Least Squares (OLS) model is utilized to identify significant variables affecting pedestrian volume. The dataset is then split into training and testing datasets at a ratio of 70:30, and 5-fold cross-validation is applied to address overfitting concerns. Subsequently, the performances of Gradient Boosting Decision Tree (GBDT), Random Forest and XGBoost models are compared and analyzed, with the model exhibiting the highest accuracy being selected.

Finally, the chosen machine learning algorithm is integrated with SHAP (SHapley Additive exPlanations) to ascertain the relative importance of variables and unravel the nonlinear threshold and interaction effects between urban design, air quality, traffic vehicles and pedestrian volume.

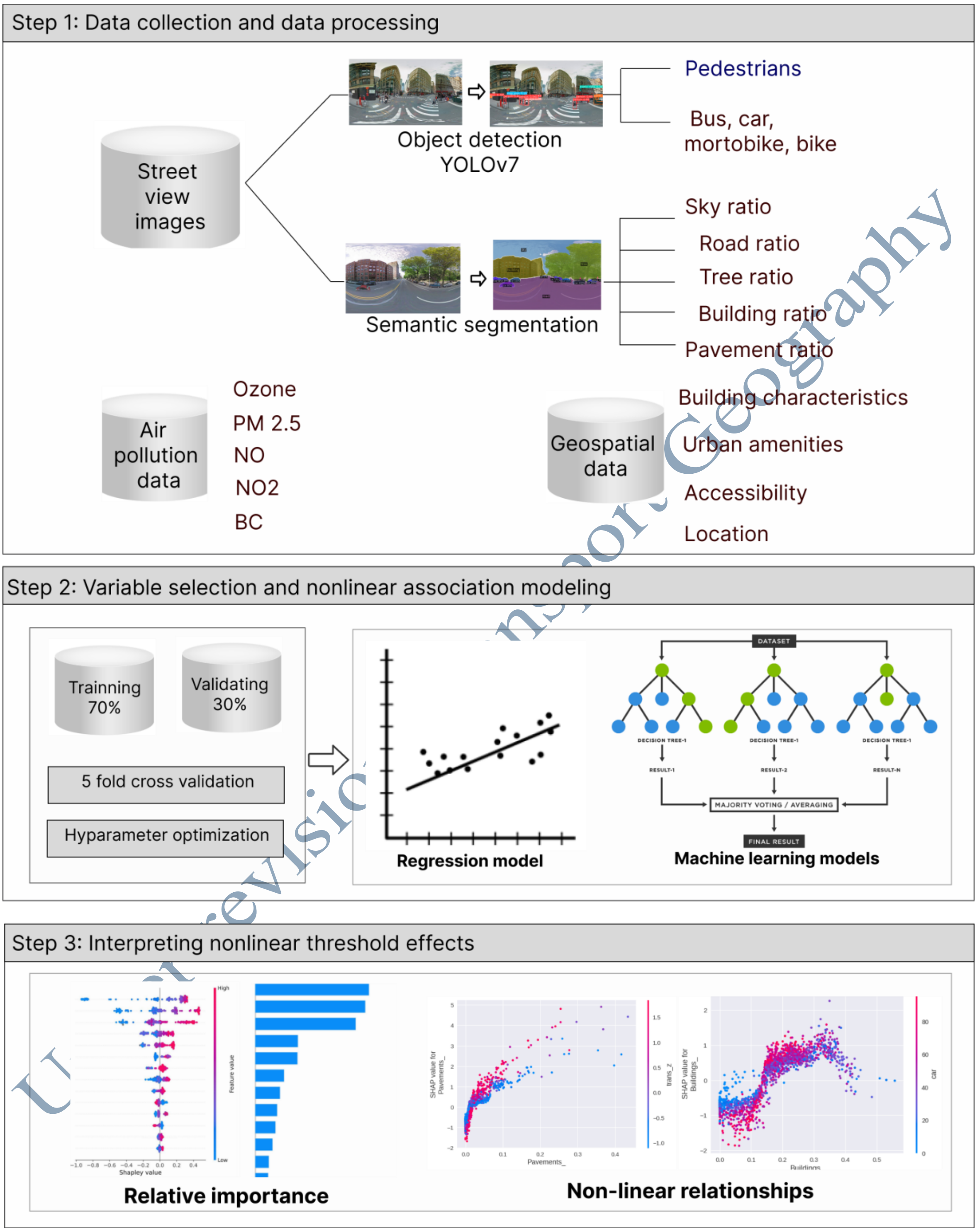

Figure 2. Analytical analysis flow

4.3. Data collection and processing

4.3.1. Pedestrian volume and traffic vehicles

In this study, the dependent variable is the number of pedestrians. Pedestrian and vehicle extraction from Street View Images (SVI) was performed using YOLOv7, a reliable technique for object detection widely utilized in recent studies (Jiang et al., 2022; Patel et al.). A minimum threshold of 0.6 was set to enhance the detection process's accuracy. Pedestrians and vehicles (bus, bike, car, motorbike) with a confidence level below 0.6 were considered undetected and excluded from the recorded count.

4.3.2 Micro-scale urban design characteristics

Previously, earlier studies relied on manual extraction of urban design features. However, recent advancements in deep learning algorithms have enabled more accurate extraction of urban streetscape design. Scholars have increasingly employed street view images and deep learning techniques to assess greenery, walkability, and sky view index (Wu et al., 2023; Yang et al., 2024a; Yang et al., 2023).

Street view images have gained popularity for reflecting street design due to their confirmed reliability (Liu et al., 2023a). In this study, we utilized 29,540 street view images of Manhattan, New York City, obtained from the Google API. These images were captured at 2.5-meter intervals using a 360° camera mounted on a vehicle, ensuring comprehensive coverage of the central street network in Manhattan. The images were taken in four directions (0°, 90°, 180°, 270°) to capture the streetscape design accurately. We applied the Xception-71 Convolutional Neural Network (CNN) to analyze the images, pre-trained on annotated images from the Cityscapes database. The Cityscapes database comprises pixel-level annotated street scenes from 50 cities, providing a robust benchmark. The Xception-71 CNN demonstrated superior performance compared to alternative CNN architectures.

We determined the average number of pixels per class per panorama for each sampling point and each object class. In this study, we focused on five streetscape features:

the rate of building, road, sky, tree, and pavement. The calculation of streetscape features for a panorama-generating position is as follows:

$$\text{Streetscape}_{\text{tree}} = \frac{pixel\,(tree)}{\sum pixcels}$$

Streetscape $_{\text{tree}}$ presents the ratio of the tree in the image, pixel(tree) presents tree pixels, $\sum pixcels$ is total of pixel in an image. The formula for Streetscape $_{\text{sky}}$, Streetscape $_{\text{building}}$, Streetscape $_{\text{road}}$, and Streetscape $_{\text{pavement}}$ is similar as Streetscape $_{\text{tree}}$.

Finally, we integrated them into 100 x 100 grid spatial analysis unit before data analysis.

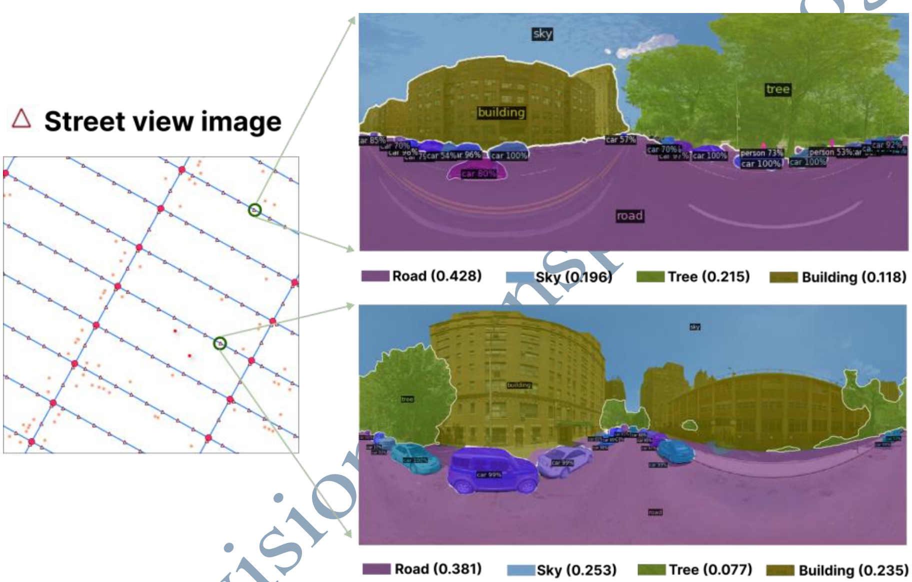

Figure 3. Semantic segmentation sample

4.3.3. Macro-scale urban design and air quality data

We collected potential neighbourhood factors that could affect pedestrian volume. The first data set pertains to Points of Interest (POI), which provide spatial and attribute information about community characteristics. POI data was gathered from Overpass Turbo, a platform offering various datasets. This study considered 11 types of POI, including education, health services, markets, businesses, theatres, parks, restaurants, galleries, banks, convenience stores, and department stores. Transportation data, such as metro and bus stations, intersection was used. Following previous studies, we included macro-scale urban design such as built floor area ratio, commercial ratio, facility floor area ratio, residential floor area ratio, number of buildings, and number of floors.

Acknowledging previous studies that highlight the impact of air pollution on pedestrian volume, we incorporated air pollution variables into our model. Air pollution data were sourced from NYC Open Data, offering raster data of the annual average predicted values of fine particulate matter ($PM_{2.5}$), nitric oxide (NO), and ozone concentration (O_3) (see Table 1).

Table 1. Descriptive variables

	Mean	Std	Min	Max
Traffic vehicles				
Motorcycle	0.19	0.62	0	22
Bicycle	0.18	0.58	0	9
Car	48.42	27.53	0.00	221.00
Bus	1.20	1.85	0.00	20.00
Macro-scale urban design				
Intersection	0.87	1.17	0.00	20.00
Number of buildings	6.45	18.12	0.00	134.00
Number of floors	8.21	7.62	0.00	104.00
Built floor area ratio	4.56	3.75	0.00	29.76
Resident floor area ratio	4.70	2.91	0.00	10.00
Commercial floor area ratio	2.46	4.04	0.00	15.00
Facility floor area ratio	6.08	3.28	0.00	15.00
Catering services#	0.06	1.01	-0.89	4.40
Shopping services#	0.08	0.98	-1.59	3.63
Residential services#	0.03	1.02	-0.63	5.04
Recreational services#	0.04	1.02	-0.61	4.49
Scientific services#	0.04	1.02	-0.76	6.57
Medical services#	0.06	1.00	-1.05	5.48
Transportation services#	0.06	1.00	-1.11	3.97
Governmental services#	0.04	1.01	-0.98	3.73
Financial services#	0.06	1.01	-0.87	4.58
Corporation services#	0.02	1.02	-0.40	12.78
Official services#	0.01	1.02	-0.24	14.29
Micro-scale urban design				
Pavement ratio (%)	0.02	0.04	0.00	0.43
Sky ratio (%)	0.26	0.09	0.00	0.61
Building ratio (%)	0.16	0.11	0.00	0.56
Tree ratio (%)	0.11	0.10	0.00	0.53
Road ratio (%)	0.35	0.05	0.03	0.58
Air quality				
Ozone concentration (ppb)	27.97	1.13	23.67	32.46

Nitric oxide (ppb)	19.83	2.83	12.47	27.54
PM 2.5 ($\mu g/m^3$)	7.37	0.81	6.19	9.68

Note: # Variables were transformed to z-score.

4.4. Methods

4.4.1. OLS and machine learning approach

Follow previous studies (Doan, 2023), we initially employed a linear regression model to ascertain the relationship between explanatory variables and pedestrian volume, aiming to identify statistically significant variables influencing walking volume. Specifically, independent variables included urban design, air pollution, and traffic vehicles, while pedestrian volume is the dependent variable. The linear regression model is presented in Eq1.

$$Y = a + \sum_{i=0}^{n} a_i . X_i + \epsilon \qquad (1)$$

Where Y is the pedestrian volume in each spatial analysis, X_i represents the independent variables (urban design, air pollution and vehicles) and a_i is the coefficient of the independent variables.

Given that spatial autocorrelation might be present among the variables, this study utilized both the spatial lag model and spatial error model to uncover spatial effects, which the OLS model may not adequately capture. The SLM and SEM are expressed (Eq. 2 to Eq. 4) as follows:

$$Y = \rho W_y + \alpha_i X_i + \epsilon \qquad (2)$$

$$Y = \alpha_i . X_i + \vartheta \qquad (3)$$

$$\vartheta = \tau . W_0 + \mu \qquad (4)$$

Where Y is pedestrian volume, α_i is the coefficient of dependent variable, X_i is the i^{th} independent variables, and ϵ is residual error; W_y is a weighted average of pedestrian volume at nearby locations, and ρ is spatial dependence parameter. ϑ is a vector of spatially autocorrelated residuals; τ is an autoregressive parameter; W_0 is a weighted average of error terms at nearby locations; μ is a vector of independent residuals.

We selected variables with VIF values below 7.5 and statistically significant at a 5% level to avoid multicollinearity before incorporating them into the machine learning models. The chosen variables serve as inputs for RF, GBDT and XGBoost models. These models, have been widely utilized in previous studies to identify non-linear relationships. Notably, XGBoost outperformed both GBDT, RF and OLS models in estimating pedestrian volume. XGBoost is a robust non-linear machine learning technique that utilizes boosting to create highly accurate predictive models (Li et al., 2024). Introduced by Chen and Guestrin (2016), it has gained popularity across various domains for its exceptional performance. The core algorithm of XGBoost is gradient boosting, an ensemble method that combines weak learners to form a more powerful learner. Each subsequent learner addresses the limitations of its predecessors, resulting in improved predictive capabilities.

4.4.2. SHAP

Interpretation ability poses a significant challenge for machine learning models, an aspect often overlooked in many studies. Despite the high prediction accuracy achieved by these models, the interpretation remains a crucial aspect. In 2017, Lundberg and Lee (2017) introduced the SHAP (SHapley Additive exPlanations) approach as a solution to interpret the outcomes of machine learning models. One of the key advantages of SHAP is its ability to unveil feature contributions in a dataset, reflecting both negative and positive influences. The SHAP method interprets machine learning models through Shapley values, leveraging concepts from game theory initially introduced by Shapley (1953) to estimate the importance of explainable variables. SHAP has been used in recent studies (Lan et al., 2023; Luo et al., 2024; Zheng et al., 2023).

5. Results

5.1. Variable selection and model performance

To identify collinearity among variables, we assessed the VIF values, eliminating variables with VIF exceeding 7.0. Subsequently, we applied the OLS stepwise model to select variables demonstrating statistical significance about pedestrian volume. Following the OLS regression, 22/30 variables remained, encompassing multiple urban design features such as pavement, sky, building ratio, macro-scale urban design, vehicles and ozone concentration (Table 2). Our findings highlight the pavement ratio as the most influential feature in promoting walking activities among the micro-scale urban design characteristics. Notably, ozone concentration significantly negatively impacted pedestrian volume, consistent with the results of and Huang et al. (2023); Jiang et al. (2019). The outcomes underscore that motorcycles, bicycles, and buses significantly impact pedestrian volume, indicating that higher numbers of bicycles and motorbikes correlate with increased pedestrian activity. Moreover, elements of the macro-scale urban design, including the number of intersections and transportation and catering services, also significantly influenced pedestrian volumes.

Table 2. Regression results

Term	OLS	SEM	SLM	VIF
	Coef. (SE)	**Coef. (SE)**	**Coef. (SE)**	
Road vehicles				
Motorcycle	0.682*** (0.136)	0.636*** (0.128)	0.659*** (0.130)	1.041
Bicycle	1.654*** (0.151)	1.498*** (0.142)	1.548*** (0.144)	1.123
Car	0.025*** (0.003)	0.030*** (0.003)	0.026*** (0.003)	1.543
Bus	0.301*** (0.049)	0.347*** (0.048)	0.321*** (0.047)	1.221
Air quality				
O_3	-0.494*** (0.092)	-0.486*** (0.122)	-0.375*** (0.088)	1.615
Micro-scale urban design				
Pavement	26.506*** (2.695)	26.232*** (2.749)	23.727*** (2.584)	1.189

Term	OLS Coef. (SE)	SEM Coef. (SE)	SLM Coef. (SE)	VIF
Sky	4.896*** (1.286)	6.717*** (1.365)	5.338*** (1.229)	1.748
Building	7.426*** (1.445)	4.206*** (1.532)	4.148*** (1.385)	3.364
Macro-scale urban design				
Intersection density	0.900*** (0.077)	0.925*** (0.075)	0.896*** (0.074)	1.198
Number of buildings	0.052*** (0.005)	0.049*** (0.007)	0.043*** (0.005)	1.632
Number of floors	-0.101*** (0.019)	-0.107*** (0.021)	-0.094*** (0.018)	3.110
Built floor area ratio	0.280*** (0.053)	0.271*** (0.056)	0.273*** (0.051)	5.924
Resident floor area ratio	-0.194*** (0.057)	-0.137** (0.069)	-0.117** (0.549)	4.093
Commercial floor area ratio	0.398*** (0.045)	0.417*** (0.054)	0.304*** (0.043)	4.828
Facility floor area ratio	0.288*** (0.064)	0.213*** (0.076)	0.191*** (0.061)	6.448
Catering service density	1.079*** (0.195)	1.096*** (0.264)	0.730*** (0.187)	5.643
Residential service density	0.799*** (0.165)	0.822*** (0.223)	0.473*** (0.158)	4.114
Recreational service density	-0.482*** (0.151)	-0.466** (0.207)	-0.321** (0.144)	3.455
Science service density	0.378*** (0.101)	0.401*** (0.140)	0.291*** (0.097)	1.558
Medical service density	0.395*** (0.134)	0.473** (0.184)	0.340*** (0.128)	2.657
Transportation service density	0.964*** (0.197)	1.231*** (0.263)	0.611*** (0.189)	5.679
Government service density	-0.937*** (0.120)	-1.031*** (0.163)	-0.733*** (0.115)	2.143
Constant	11.787*** (2.511)	11.636*** (3.355)	7.643*** (2.410)	
ρ / τ		0.328*** (0.016)	0.301*** (0.015)	
R^2	0.472	0.520	0.516	
AIC	33966.3	33606.9	33633.4	

*Note: *** Significant at 99% confidence level; ** Significant at 95% confidence level*

Figure 4 depicts the global Moran's I of pedestrian volume, yielding a Moran's I value of 0.510, signifying a significant and positive spatial autocorrelation. This observation aligns with findings from existing studies (Chen et al., 2022a; Lee et al., 2017). In Table 2, the results of SEM and SLM (columns 3 and 4) showcase the estimated relationships between pedestrian volume and vehicles, air quality and urban design. The outcomes indicate that the spatial autocorrelation parameters are significant, with positive coefficients in both models, which aligns with our expectations. The SEM and SLM models perform better than their non-spatial counterparts, evidenced by higher R^2 and lower AIC values. These results confirm the positive spatial autocorrelation for pedestrian volume. This observation aligns with the findings of Chen et al. (2022a). However, we observed a nonlinear residual distribution shape. Consequently, we employed nonlinear machine learning models in the subsequent steps to explore the complex association between urban design, air quality, vehicles and pedestrian volume.

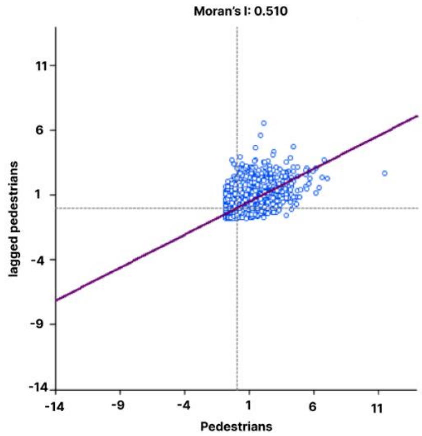

Figure 4. Moran's I of pedestrian volume

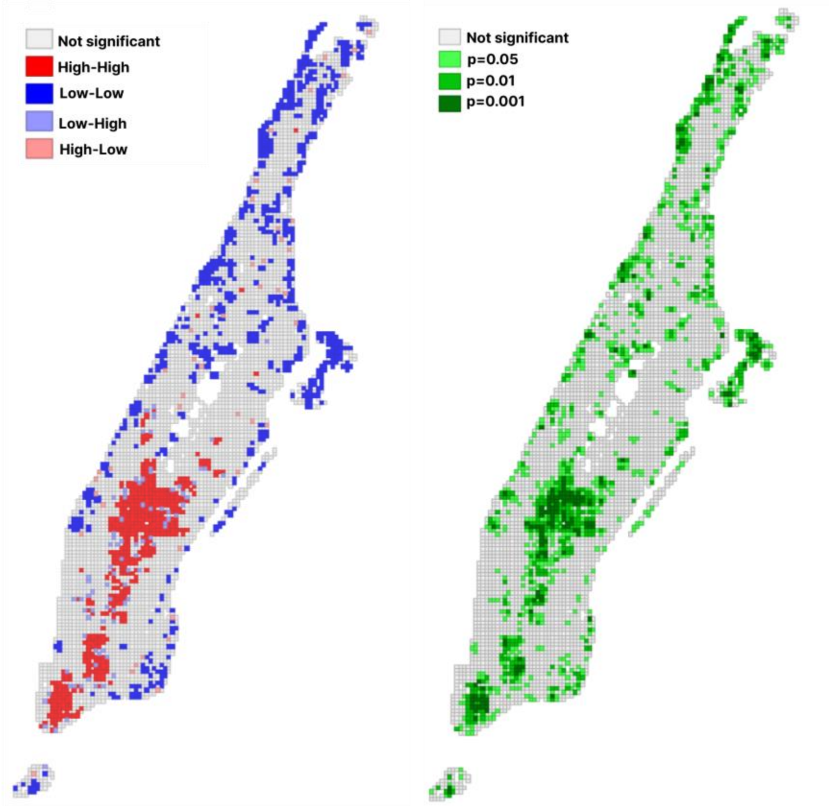

Figure 5. LISA cluster (left) and LISA significant (right) of pedestrian volume

We employed three machine learning models (RF, GBDT, and XGBoost) and compared their performance in predicting pedestrian volume. The machine learning models demonstrated significantly better performance than the linear regression model, with improvements ranging from 9.95% to 13.81%. XGBoost exhibited the highest predictive power at 0.5304, surpassing RF and GBDT by 3.5% and 0.7%, respectively. Regarding other evaluation indicators, XGBoost demonstrated the smallest prediction error in the training and validation datasets. The MAE of RF and GBDT was 3.7409 and 3.7300, respectively, compared to 3.6826 for XGBoost. Similarly, the MSE of XGBoost was 34.5323, slightly lower than RF and GBDT. While the RMSE of RF and GBDT stood at 5.9876 and 5.8966, respectively, XGBoost achieved a lower figure of 5.8764. Consequently, XGBoost was utilized to uncover the relative importance, nonlinearity, and threshold identification between urban design, air pollution, vehicles, and pedestrian volume.

Table 3. Model evaluation

Model	MAE	MSE	RMSE	R^2	The improvement compared with OLS
RF	3.7409	35.8515	5.9876	0.5124	9.95%
GBDT	3.7300	34.7707	5.8966	0.5271	13.11%
XGBoost	3.6826	34.5323	5.8764	0.5304	13.81%

5.2. Relative importance of variables to pedestrian volume

Figure 6 shows the relative importance of explanatory variables. Each row represents a feature and is ranked based on the mean |SHAP value|. The results demonstrate that urban micro-scale urban design significantly contributes to pedestrian volume estimation, accounting for 14.26%. Specifically, pavement and building ratios rank as the fifth and sixth most important features, contributing 6.16% each. With less importance, the contribution of the sky ratio is 1.94%, much lower than that of the pavement and building ratio. Our findings align with the observations of Chen et al. (2022a), who pointed out that sidewalks and open sky are important factors in promoting pedestrian volume among micro-scale urban design features.

As depicted in Fig 6, vehicles contribute 18.57% to the influence on pedestrian volumes, with the contributions of cars and bicycles at 5.94% and 5.72%, respectively. Among the four modes of transport, buses significantly influence pedestrian volume, accounting for 4.75%, compared to only 2.16% for motorcycles. Our findings underscore the importance of public vehicles on the street in promoting walking activities. Although the contribution of cars to pedestrian volumes is 5.94%, we suggest that a car-based approach to promoting pedestrian volume is a double-edged sword. On the one hand, cars might increase accessibility and mobility for people. On the other hand, overcrowded cars can lead to social issues such as traffic congestion and air pollution (Adams and Requia, 2017; Yasin et al., 2012), indirectly influencing pedestrian activities. Therefore, a bicycle-based approach should be considered to encourage people to walk.

Additionally, transportation services have the most substantial influence on pedestrian volume. This finding aligns with numerous previous studies on walking activity and its determinants (Hajrasouliha and Yin, 2014; Jiao et al., 2021; Liu et al., 2023b), who emphasized the important influence of accessibility and connectivity on physical activity. Notably, transportation services can predict 18.47% of the variance in pedestrian volume. They connect urban spaces, and reasonable transportation services such as buses, metro, and railways create a convenient environment for people to walk (Jiang et al., 2021). The importance of catering services in impacting pedestrian volume is ranked second, contributing 10.15%. The commercial floor area ratio is a significant feature in predicting pedestrian volume, with 8.64%, followed by the number of intersections. These findings support Jacobs' theory on urban vibrancy (Fuller and Moore, 2017). We also observed that the number of intersections significantly contributes to walking volume, accounting for 8.21% of pedestrian volume explanation. Furthermore, air pollution contributes 2.81% to the overall influence on pedestrian volume.

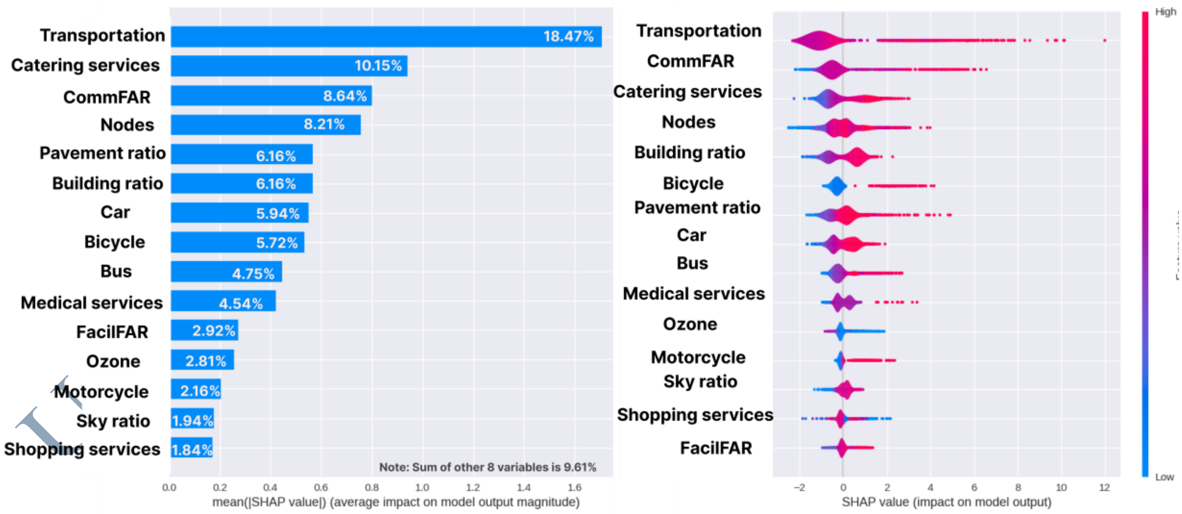

Figure 6. Relative importance of variables

5.3. Nonlinear relationship and threshold identification

In addition to their relative importance, the SHAP model reveals the nonlinear relationship and threshold identification between dependent and independent variables. The dependence SHAP plot can visualize the marginal effects of the explanatory variables on the target variable. In this part, we present plots of the most important features in each group, including streetscape design (pavement, building, sky ratio), ozone pollution, traffic vehicles (bus, car, motorbike, bicycle), and urban facilities (transportation, catering and commercial services, interaction).

Figure 7 demonstrates nonlinear relationships between micro-scale urban design and pedestrian volume. The pavement ratio positively influences pedestrian volume and follows a nonlinear association (fig.7a). When the pavement ratio exceeds 0.05, its impact on pedestrian volume increases as the SHAP value rises. This is because wide sidewalks can create a safe and pleasant walking environment and thus promote walking behaviours (Nagata et al., 2020). Similarly, the building ratio exhibits a clear nonlinear relationship with pedestrian volume (fig.7b). When the building ratio is under 0.1, the impact on pedestrian volume is stable. The pedestrian volume increases significantly as the building ratio increases from 0.1 to 0.33, aligning with the study of Kang (2018). Beyond the threshold of 0.33, the influence of the building ratio shifts from positive to negative. The possible explanation for reducing pedestrian volume once the building ratio exceeds 0.33 is a less attractive environment for walking, as explained in previous studies (Böcker et al., 2017; Cheng et al., 2020; Liu et al., 2023b). The nonlinear relationship between sky ratio and pedestrian volume is illustrated in Fig. 7c, indicating a significant positive impact as the sky ratio increases from 0.1 to 0.35. This finding is supported by studies of Wang et al. (2019) and Chen et al. (2022a), who observed visual enclosure (sky ratio) positive impact on pedestrian volume. However, the pedestrian volume shows minor differences if the sky ratio exceeds the threshold of 0.35. Our findings are inconsistent with the finding of Yin and Wang (2016), who observed a negative association between open sky and pedestrian volume. The linear assumption in the study of Yin and Wang (2016) may explain this inconsistency. The interaction between the sky ratio and catering services shows that a high ratio of the sky cannot further increase pedestrian volume if limited catering services are provided. This finding emphasizes the importance of integrating streetscape design elements and urban facilities to create attractive walking environments. A greater choice of urban amenities and service destinations can better stimulate walking behaviour (Hahm et al., 2017).

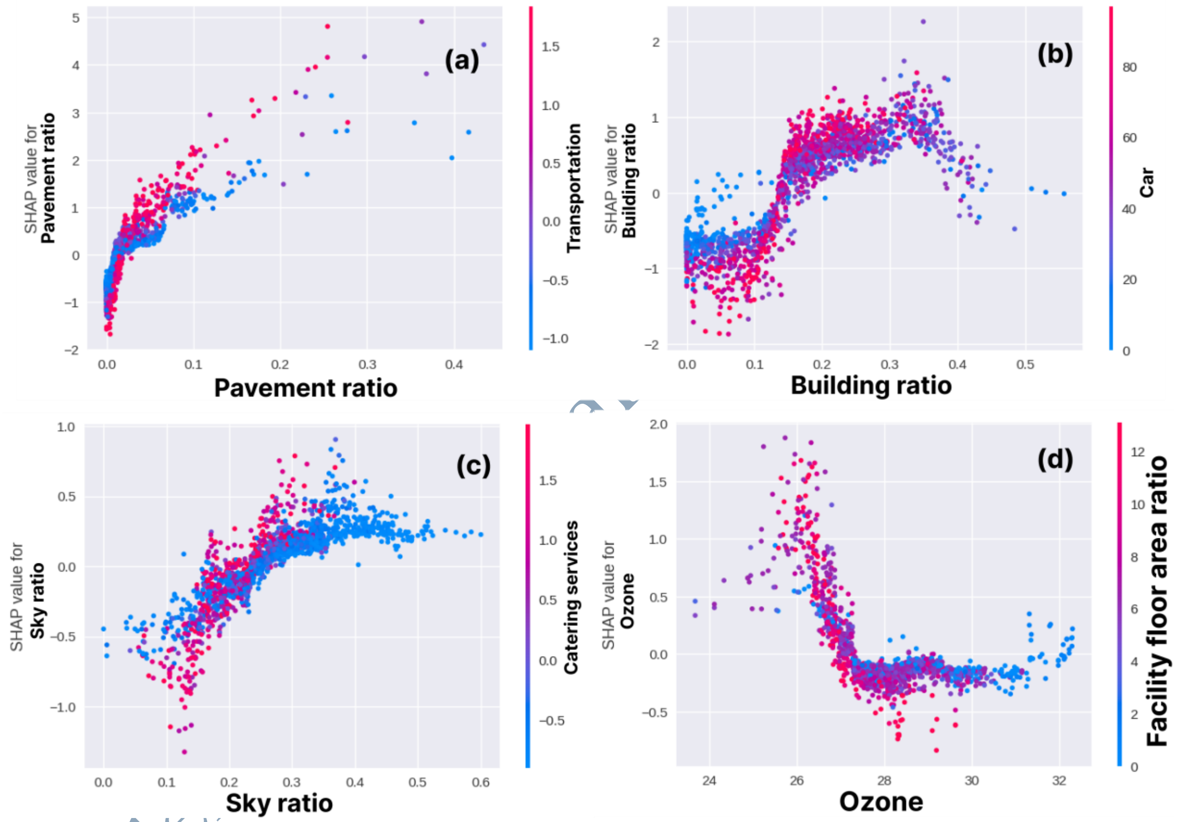

Figure 7. Nonlinear and interaction effects of micro-scale urban design, ozone concentration on pedestrian volume

Using OLS models, previous studies indicated that air pollution negatively influences pedestrian activities. Employing nonlinear machine learning algorithms, we observed that the impact of air pollution on pedestrian volume does not follow the linear assumption (fig. 7d). In general, the impact of air pollution on pedestrian volume is negative. Specifically, the pedestrian volume decreases significantly when the ozone concentration increases to 27.3 ppb. Low air quality can affect pedestrians' satisfaction and their experiences (Ma et al., 2021). With high pollution levels, people will tend to use other kinds of transportation as avoidance behaviours (Zhao et al., 2018).

Often neglected in previous quantitative studies, we have identified that vehicles exert significant nonlinear and threshold influences on pedestrian volume (figure 8). The pedestrian volume increases significantly for cars within the range of 30 to 60; however, beyond this range, the influence becomes trivial (fig. 8a). Similar patterns are observed for bicycles and motorbikes, where having more than two of each positively influences pedestrians (fig. 8b-c). The Shapley value corresponding to the number of buses depicts a nonlinear relationship and a threshold effect, as shown in Fig. 8d. Within the range of 0-5, the pedestrian volume increases linearly, but beyond this range, the effect becomes trivial as the Shapley value slightly decreases.

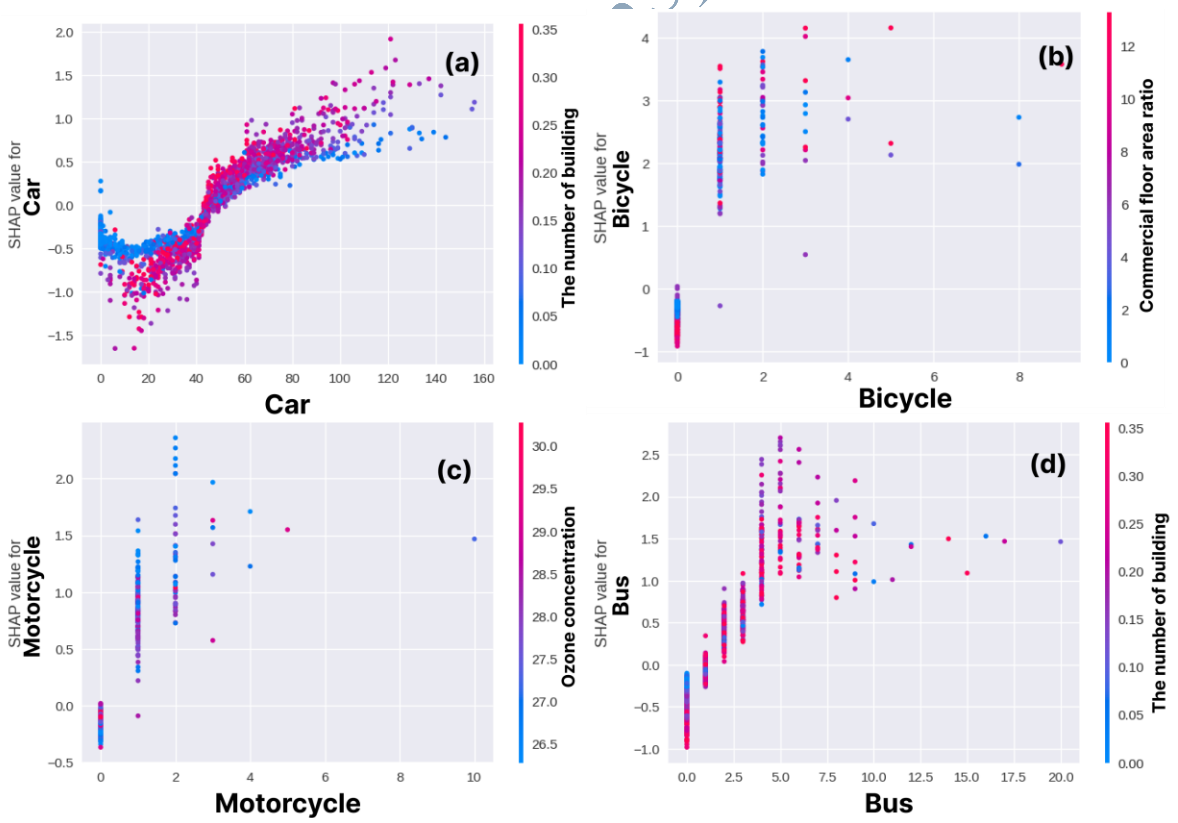

Figure 8. Nonlinear and interaction effects of vehicles on pedestrian volume

The associations between macro-scale urban design features and pedestrian volume are illustrated in Figure 9. The local effects significantly increase once transportation service exceeds about 1 (Figure 9a). This phenomenon aligns with previous studies indicating that moderate street connectivity provides convenient access for walkers (Hajrasouliha and Yin, 2014). Additionally, a higher density of roads and transportation services (bus stops, stations) may correspond to dense people flow and traffic, potentially hindering pedestrian volume (Chen and Ye, 2021; Huang et al., 2022). Similarly, catering services positively affect pedestrian flow (Figure 9b). The local effects transition from negative to positive when catering service density exceeds 0.5. This finding aligns with studies by and Xia et al. (2020); Ye et al. (2018), who argue that catering services represent a vibrant place that can attract people to walk. The commercial floor area ratio (ComFAR) positively affects pedestrian volume beyond a threshold of 6, implying that pedestrian volume increases as the ComFAR increases. Furthermore, figure 9c shows that a higher commercial floor area ratio and high catering services lead to higher pedestrian volume. This is because areas with a mix of commercial activities and catering services provide opportunities for social interaction. People may gather in these places for meals, meetings, or socializing, contributing to a vibrant and lively environment that attracts pedestrians (Cerin et al., 2007; Duncan et al., 2010). The number of intersections significantly affects pedestrian volume. The impact is relatively light when the variable exceeds 5 intersections/grid (fig. 9d). This observation aligns with previous studies by and Chen et al. (2022a); Hajrasouliha and Yin (2014); Jiang et al. (2021), arguing that street connectivity provides pedestrians convenient access to urban amenities.

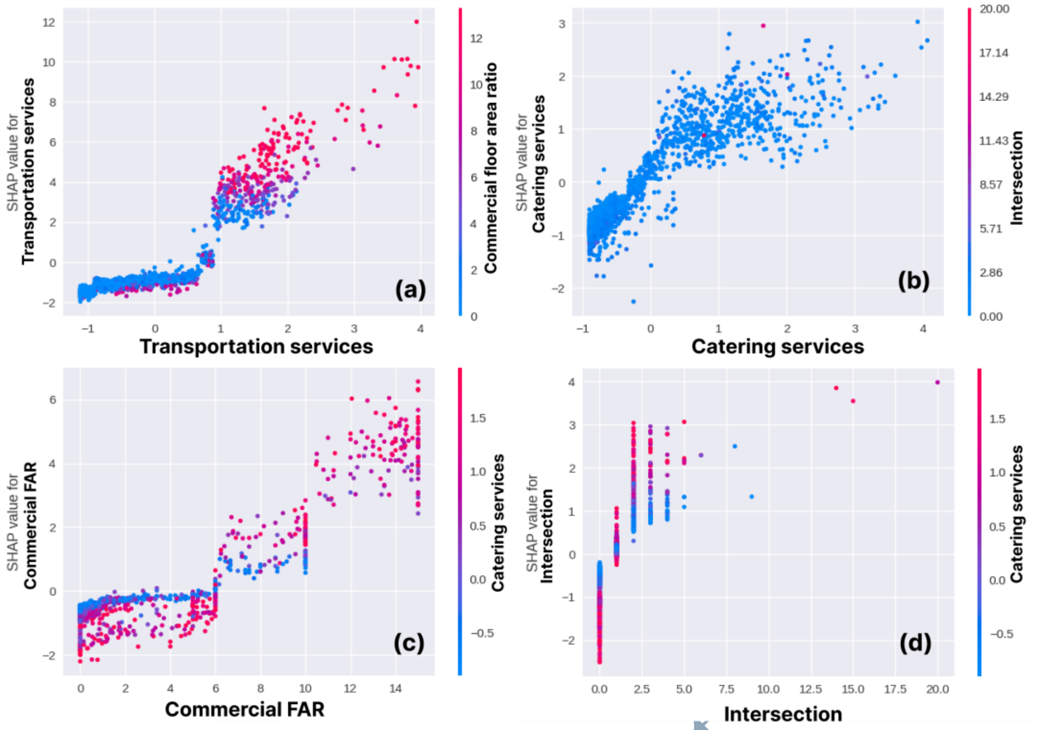

Figure 9. Nonlinear and interaction effects of macro-scale urban design features on pedestrian volume

6. Discussion

6.1. Importance role of urban design, vehicles and pollution

This study employed explainable machine learning to uncover the nonlinear threshold effects of multiple-scale urban design, traffic vehicles, and air quality on pedestrian volume. The results indicate that macro-scale urban design has a more pronounced influence on walking activity, aligning with the hierarchy of walking needs proposed by Alfonzo (2005). Among the macro-level urban design factors, transportation and catering services emerged as the most crucial, accounting for 18.47% and 10.15%, respectively, in affecting pedestrian volume. Regarding micro-scale urban design, pavement and building ratios were identified as the most important determinants, contributing 6.16% each. These findings are consistent with the studies of Wang et al. (2022); and Wu et al. (2023), highlighting the significance of factors like enclosure in relation to walking activities.

Notably, the relative importance ranking of road vehicles, including cars, bicycles, and buses, is emphasized in this study, which has been lacking in previous research. While cars and bicycles contribute 5.49% and 5.72%, respectively, to predictive power, the relative importance of buses is 4.75%. This observation aligns with previous studies that revealed the role of public transport in encouraging people to walk instead of private vehicles (Freeland et al., 2013; Panter et al., 2016). Furthermore, the study found that air pollution contributes 2.81% to pedestrian volume prediction when considering air quality within the urban design context. Although the impact of air quality on pedestrian volume is relatively minor, it surpasses the importance of several urban design features. Therefore, the influence of air quality on walking activities should not be overlooked (Tainio et al., 2016; Tainio et al., 2021).

6.2. Nonlinear and interaction effects of urban design, vehicle and air quality on pedestrian volume

Our findings partially align with existing studies, revealing a nonlinear relationship between urban design and pedestrian activity. Notably, we discovered that enhancing the pavement ratio is a key factor in promoting walking activities among micro-level urban design features. This result is consistent with prior studies, which have identified that well-designed pavements contribute to a connected urban network, offering pedestrians direct and convenient routes between destinations. Our study also identified interactions between pavement ratio, transportation services, sky ratio, and catering services. These interactions underscore the importance of integrating different aspects of urban design. Collaborating urban amenities and landscape elements can enhance the aesthetic appeal of public spaces, making them more inviting for pedestrians (Karusisi et al., 2012). Additionally, micro-scale urban design elements, such as building ratio, play a crucial role in promoting walking activities, aligning with findings from previous studies (Chen et al., 2022a; Kaplan et al., 2016; Koo et al., 2021). However, our study revealed a significant impact of building ratio on pedestrian volume within the range of 0.15 to 0.35. Exceeding a ratio of 0.35, the local effects transition from positive to negative. This shift could be attributed to the 'canyon effect' created by closely spaced tall buildings, which may make pedestrians feel enclosed, leading to a sense of congestion, reduced visibility, and a less inviting walking environment. This observation resonates with studies by Cheng et al. (2020) and Yang et al. (2024b), emphasizing that densely built areas might not be conducive to comfortable physical activities such as walking and jogging. Sky ratio also exhibits a positive nonlinear relationship with pedestrian volume. This point is consistent with the confirmed hypothesis that open areas can attract more walkers because of reduced pressure (Yang et al., 2023).

In terms of macro-scale urban design, our study reaffirms that transportation services, such as bus stops and stations, significantly influence pedestrian volume, consistent with findings from studies by (Guo et al., 2023; Liu et al., 2021; Zhou et al., 2024). However, we employed XGBoost and SHAP to identify a threshold effect in the relationship between transportation services and pedestrian volume. Specifically, when transportation service surpasses a threshold of 1.0, pedestrian volume experiences a significant increase. Notably, we observed an interaction between transportation services and commercial floor area ratio (CommFAR) – when CommFAR exceeds 6 and transportation services are higher than 1, pedestrian volumes are extensively promoted. This finding suggests a collaborative promotion of pedestrian activity in specific ranges by these two factors. Our study emphasizes the importance of providing convenient transportation for commercial areas to enhance urban accessibility and foster social interaction. The number of intersections exhibits a positive nonlinear relationship with pedestrian volume, aligning with previous studies (Yang et al., 2021a; Yang et al., 2024b). Interestingly, our study identified an optimal intersection density of 4 intersections per grid, beyond which further increases do not result in a corresponding rise in pedestrian volume. Moreover, intriguing synergistic effects were observed between the number of intersections and catering services – grids with higher catering services attracted much more pedestrian volume, holding the number of intersections constant.

We identified nonlinear threshold effects of vehicles and air quality on pedestrian volume. While road vehicles exhibit a positive relationship with pedestrian volume, the negative impact of private vehicles on environmental issues should be considered. Pedestrian volume remains stable when ozone concentration exceeds 27.3 ppb, inconsistent with previous studies that highlight the negative impact of air pollution on physical activities. In the context of Manhattan, we hypothesize that there is a trades-off between pollution exposure and walking on the street. In other words, individuals may choose between enjoying the urban environment and the potential health risks associated with increased pollution exposure. To clarify this hypothesis, future empirical research will be needed to validate or refute the idea of a trade-off between pollution exposure and participating in activities in the context of Manhattan, NYC.

6.3. Sustainable urban design policy implications

This study provides insights into urban design and planning, suggesting several targeted interventions. Firstly, the hierarchy of intervention priorities can be determined by considering factors such as transportation services, catering facilities, commercial floor area ratio, intersections, pavement quality, building ratio, and types of road vehicles (cars, buses, bicycles). Beyond the urban design aspect, strategies for managing air quality should also be considered, as ozone concentrations contribute 2.18% to pedestrian volume estimation. While walkers may be less sensitive to air quality at lower pollution levels, an increase beyond a threshold can impact human health, prompting avoidance behaviors and potentially reducing pedestrian volume. Second, the nonlinear and threshold identification of urban design on pedestrian volume provides more effective measures to promote walking activities. For instance, effective ranges for the building ratio should be set between 0.15 and 0.35, and the sky ratio and transportation services should be maintained above 0.3 and 1, respectively. Controlling building density and height within a reasonable range can help maintain good street openness and an attractive design landscape (Tabatabaie et al., 2023), thus promoting walking. Additionally, among various modes of transportation, cars and bikes are the most substantial contributors, followed by buses. It's worth noting that the contribution of vehicles is often overlooked in pedestrian determinant studies. While the number of cars positively influences pedestrian volume, urban managers should consider the dual influences of private vehicles. Third, identifying interaction effects suggests that authorities' interventions to promote pedestrians should focus on multiple dimensions. For example, urban planners should consider transportation services and commercial floor area ratios. Catering services and sky ratio also contribute to constructing an attractive walking environment.

6.4. Limitation and future works

While this study provides insightful findings, it is not without limitations. Firstly, the results may be context-specific, and comparative studies across different cities are recommended to draw more general conclusions. It would be beneficial to compare financial centres within contexts similar to those in Manhattan, NYC. Secondly, urban design's identified nonlinear threshold effects may not necessarily imply causal relationships. Future studies should conduct longitudinal experiments to uncover the causal mechanisms behind these effects. Thirdly, although the street view images utilized in this study can extract pedestrian flows and vehicle volume on spatial dimensions, capturing temporal units remains challenging. In future research, integrating other urban big data sources, such as mobile phone and trajectory data, could be explored to unveil the temporal relationship between urban design, pollution, vehicle flow and pedestrian volume.

7. Conclusion

This study unveils the nonlinear effects of urban design on pedestrian volume, using the case of Manhattan, NYC, based on urban big data and explainable machine learning. We identified the critical urban design factors influencing pedestrian volume and revealed nonlinear threshold effects among urban design, vehicles, air quality, and pedestrian volume. The main findings include: (1) Macro-scale urban design plays a more important role than micro-scale urban design in determining pedestrian volume, with transportation services and catering services being the two most significant contributors, followed by commercial area ratio. Moreover, vehicles (i.e., buses and bikes) significantly promote pedestrian volume. Air quality contributes 2.18% to pedestrian volume estimation. (2) All urban design elements, vehicles, and air quality exhibit nonlinear effects on pedestrian volume. (3) Interactions between explanatory variables suggest that urban planning should consider multiple dimensions. These findings have important implications for urban planners and designers to formulate policies and guidelines for designing a more sustainable urban environment for pedestrians.

8. Reference

Adams, M.D., Requia, W.J. 2017. How private vehicle use increases ambient air pollution concentrations at schools during the morning drop-off of children. *Atmospheric environment*, *165*, 264-273.

Ahmadipour, F., Mamdoohi, A.R., Wulf-Holger, A. 2021. Impact of built environment on walking in the case of Tehran, Iran. *Journal of Transport & Health*, *22*, 101083, doi:https://doi.org/10.1016/j.jth.2021.101083.

Alfonzo, M.A. 2005. To walk or not to walk? The hierarchy of walking needs. *Environment and behavior*, *37*, 808-836.

Barnett, D.W., Barnett, A., Nathan, A., Van Cauwenberg, J., Cerin, E., Council on, E., Physical Activity –Older Adults working, g. 2017. Built environmental correlates of older adults' total physical activity and walking: a systematic review and meta-analysis. *International journal of behavioral nutrition and physical activity*, *14*, 1-24.

Böcker, L., van Amen, P., Helbich, M. 2017. Elderly travel frequencies and transport mode choices in Greater Rotterdam, the Netherlands. *Transportation*, *44*, 831-852.

Cao, J., Tao, T. 2023. Using machine-learning models to understand nonlinear relationships between land use and travel. *Transportation Research Part D: Transport and Environment*, *123*, 103930, doi:https://doi.org/10.1016/j.trd.2023.103930.

Cerin, E., Leslie, E., du Toit, L., Owen, N., Frank, L.D. 2007. Destinations that matter: associations with walking for transport. *Health & place*, *13*, 713-724.

Chen, E., Ye, Z. 2021. Identifying the nonlinear relationship between free-floating bike sharing usage and built environment. *Journal of Cleaner Production*, *280*, 124281.

Chen, L., Lu, Y., Sheng, Q., Ye, Y., Wang, R., Liu, Y. 2020a. Estimating pedestrian volume using Street View images: A large-scale validation test. *Computers, Environment and Urban Systems*, *81*, 101481, doi:https://doi.org/10.1016/j.compenvurbsys.2020.101481.

Chen, L., Lu, Y., Ye, Y., Xiao, Y., Yang, L. 2022a. Examining the association between the built environment and pedestrian volume using street view images. *Cities*, *127*, 103734, doi:https://doi.org/10.1016/j.cities.2022.103734.

Chen, S., Bao, Z., Chen, J., Yang, L., Lou, V. 2022b. Sustainable built environment for facilitating public health of older adults: Evidence from Hong Kong. *Sustainable Development*, *30*, 1086-1098, doi:https://doi.org/10.1002/sd.2303.

Chen, T., Guestrin, C., 2016. XGBoost: A Scalable Tree Boosting System, Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. Association for Computing Machinery, San Francisco, California, USA, pp. 785–794.

Chen, Y., Chen, Y., Tu, W., Zeng, X. 2020b. Is eye-level greening associated with the use of dockless shared bicycles? *Urban forestry & urban greening*, *51*, 126690.

Cheng, L., De Vos, J., Zhao, P., Yang, M., Witlox, F. 2020. Examining non-linear built environment effects on elderly's walking: A random forest approach. *Transportation Research Part D: Transport and Environment*, *88*, 102552, doi:https://doi.org/10.1016/j.trd.2020.102552.

Cheng, L., Jin, T., Wang, K., Lee, Y., Witlox, F. 2022. Promoting the integrated use of bikeshare and metro: A focus on the nonlinearity of built environment effects. *Multimodal Transportation*, *1*, 100004, doi:https://doi.org/10.1016/j.multra.2022.100004.

Chung, J., Kim, S.-N., Kim, H., 2019. The Impact of PM10 Levels on Pedestrian Volume: Findings from Streets in Seoul, South Korea, International Journal of Environmental Research and Public Health.

Doan, Q.C. 2023. Determining the optimal land valuation model: A case study of Hanoi, Vietnam. *Land Use Policy*, *127*, 106578, doi:https://doi.org/10.1016/j.landusepol.2023.106578.

Doan, Q.C., Chen, C., He, S., Zhang, X. 2024. How urban air quality affects land values: Exploring non-linear and threshold mechanism using explainable artificial intelligence. *Journal of Cleaner Production*, *434*, 140340, doi:https://doi.org/10.1016/j.jclepro.2023.140340.

Doan, Q.C., Ma, J., Chen, S., Zhang, X. 2025. Nonlinear and threshold effects of the built environment, road vehicles and air pollution on urban vitality. *Landscape and Urban Planning*, *253*, 105204, doi:https://doi.org/10.1016/j.landurbplan.2024.105204.

Duan, Z., Zhao, H., Li, Z. 2023. Non-linear effects of built environment and socio-demographics on activity space. *Journal of Transport Geography*, *111*, 103671, doi:https://doi.org/10.1016/j.jtrangeo.2023.103671.

Duncan, M.J., Winkler, E., Sugiyama, T., Cerin, E., Dutoit, L., Leslie, E., Owen, N. 2010. Relationships of land use mix with walking for transport: do land uses and geographical scale matter? *Journal of urban health*, *87*, 782-795.

Forsyth, A., Oakes, J.M., Schmitz, K.H., Hearst, M. 2007. Does residential density increase walking and other physical activity? *Urban studies*, *44*, 679-697.

Foster, C., Kelly, P., Reid, H.A.B., Roberts, N., Murtagh, E.M., Humphreys, D.K., Panter, J., Milton, K. 2018. What works to promote walking at the population level? A systematic review. *British journal of sports medicine*, *52*, 807-812.

Freeland, A.L., Banerjee, S.N., Dannenberg, A.L., Wendel, A.M. 2013. Walking Associated With Public Transit: Moving Toward Increased Physical Activity in the United States. *American Journal of Public Health*, *103*, 536-542, doi:10.2105/AJPH.2012.300912.

Fuller, M., Moore, R., 2017. An Analysis of Jane Jacobs's The Death and Life of Great American Cities. Macat Library,

Gao, J., Kamphuis, C.B.M., Ettema, D., Helbich, M. 2019. Longitudinal changes in transport-related and recreational walking: The role of life events. *Transportation research part D: transport and environment*, *77*, 243-251.

Gao, J., Kamphuis, C.B.M., Helbich, M., Ettema, D. 2020. What is 'neighborhood walkability'? How the built environment differently correlates with walking for different purposes and with walking on weekdays and weekends. *Journal of Transport Geography*, *88*, 102860, doi:https://doi.org/10.1016/j.jtrangeo.2020.102860.

Giallouros, G., Kouis, P., Papatheodorou, S.I., Woodcock, J., Tainio, M. 2020. The long-term impact of restricting cycling and walking during high air pollution days on all-cause mortality: Health impact Assessment study. *Environment International*, *140*, 105679, doi:https://doi.org/10.1016/j.envint.2020.105679.

Guo, C., Jiang, Y., Qiao, R., Zhao, J., Weng, J., Chen, Y. 2023. The nonlinear relationship between the active travel behavior of older adults and built environments: A comparison between an inner-city area and a suburban area. *Sustainable Cities and Society*, *99*, 104961, doi:https://doi.org/10.1016/j.scs.2023.104961.

Guzman, L.A., Peña, J., Carrasco, J.A. 2020. Assessing the role of the built environment and sociodemographic characteristics on walking travel distances in Bogotá. *Journal of Transport Geography*, *88*, 102844, doi:https://doi.org/10.1016/j.jtrangeo.2020.102844.

Hahm, Y., Yoon, H., Jung, D., Kwon, H. 2017. Do built environments affect pedestrians' choices of walking routes in retail districts? A study with GPS experiments in Hongdae retail district in Seoul, South Korea. *Habitat international*, *70*, 50-60.

Hajrasouliha, A., Yin, L. 2014. The impact of street network connectivity on pedestrian volume. *Urban Studies*, *52*, 2483-2497, doi:10.1177/0042098014544763.

Handy, S.L., Boarnet, M.G., Ewing, R., Killingsworth, R.E. 2002. How the built environment affects physical activity: Views from urban planning. *American Journal of Preventive Medicine*, *23*, 64-73, doi:https://doi.org/10.1016/S0749-3797(02)00475-0.

He, D., Lu, Y., Xie, B., Helbich, M. 2021. Large-scale greenway intervention promotes walking behaviors: A natural experiment in China. *Transportation research part D: transport and environment*, *101*, 103095.

Huang, D., Jiang, B., Yuan, L. 2022. Analyzing the effects of nature exposure on perceived satisfaction with running routes: An activity path-based measure approach. *Urban Forestry & Urban Greening*, *68*, 127480.

Huang, G., Jiang, Y., Zhou, W., Pickett, S.T.A., Fisher, B. 2023. The impact of air pollution on behavior changes and outdoor recreation in Chinese cities. *Landscape and Urban Planning*, *234*, 104727, doi:https://doi.org/10.1016/j.landurbplan.2023.104727.

Huber, S., 2021. Synthetization of bicycle route data from aggregate GPS-based cycling data and its utility for bicycle route choice analysis, 2021 7th International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS), pp. 1-6.

Ji, S., Wang, X., Lyu, T., Liu, X., Wang, Y., Heinen, E., Sun, Z. 2022. Understanding cycling distance according to the prediction of the XGBoost and the interpretation of SHAP: A non-linear and interaction effect analysis. *Journal of Transport Geography*, *103*, 103414.

Jiang, K., Xie, T., Yan, R., Wen, X., Li, D., Jiang, H., Jiang, N., Feng, L., Duan, X., Wang, J., 2022. An Attention Mechanism-Improved YOLOv7 Object Detection Algorithm for Hemp Duck Count Estimation, Agriculture.

Jiang, Y., Chen, L., Grekousis, G., Xiao, Y., Ye, Y., Lu, Y. 2021. Spatial disparity of individual and collective walking behaviors: A new theoretical framework. *Transportation research part D: transport and environment*, *101*, 103096.

Jiang, Y., Huang, G., Fisher, B. 2019. Air quality, human behavior and urban park visit: A case study in Beijing. *Journal of Cleaner Production*, *240*, 118000, doi:https://doi.org/10.1016/j.jclepro.2019.118000.

Jiao, J., Rollo, J., Fu, B., Liu, C. 2021. Exploring Effective Built Environment Factors for Evaluating Pedestrian Volume in High-Density Areas: A New Finding for the Central Business District in Melbourne, Australia. *Land*, *10*, 655.

Kang, C.-D. 2018. The S+ 5Ds: Spatial access to pedestrian environments and walking in Seoul, Korea. *Cities*, *77*, 130-141.

Kaplan, S., Nielsen, T.A.S., Prato, C.G. 2016. Walking, cycling and the urban form: A Heckman selection model of active travel mode and distance by young adolescents. *Transportation research part D: transport and environment*, *44*, 55-65.

Karusisi, N., Bean, K., Oppert, J.-M., Pannier, B., Chaix, B. 2012. Multiple dimensions of residential environments, neighborhood experiences, and jogging behavior in the RECORD Study. *Preventive medicine*, *55*, 50-55.

Khan, A.N. 2023. Elucidating the effects of environmental consciousness and environmental attitude on green travel behavior: Moderating role of green self-efficacy. *Sustainable Development*, *n/a*, doi:https://doi.org/10.1002/sd.2771.

Ki, D., Lee, S. 2021. Analyzing the effects of Green View Index of neighborhood streets on walking time using Google Street View and deep learning. *Landscape and Urban Planning*, *205*, 103920, doi:https://doi.org/10.1016/j.landurbplan.2020.103920.

Kim, S., Park, S., Jang, K. 2019. Spatially-varying effects of built environment determinants on walking. *Transportation Research Part A: Policy and Practice*, *123*, 188-199, doi:https://doi.org/10.1016/j.tra.2019.02.003.

Koo, B.W., Guhathakurta, S., Botchwey, N. 2021. How are Neighborhood and Street-Level Walkability Factors Associated with Walking Behaviors? A Big Data Approach Using Street View Images. *Environment and Behavior*, *54*, 211-241, doi:10.1177/00139165211014609.

Lan, H., Hou, H., Gou, Z. 2023. A machine learning led investigation to understand individual difference and the human-environment interactive effect on classroom thermal comfort. *Building and Environment*, *236*, 110259, doi:https://doi.org/10.1016/j.buildenv.2023.110259.

Lee, M., Kim, S., Kim, H., Hwang, S. 2023. Pedestrian visual satisfaction and dissatisfaction toward physical components of the walking environment based on types, characteristics, and combinations. *Building and Environment*, *244*, 110776, doi:https://doi.org/10.1016/j.buildenv.2023.110776.

Lee, S., Sung, H., Woo, A. 2017. The spatial variations of relationship between built environment and pedestrian volume: Focused on the 2009 Seoul Pedestrian flow survey in Korea. *Journal of Asian Architecture and Building Engineering*, *16*, 147-154.

Li, F., Fisher, K.J., Brownson, R.C., Bosworth, M. 2005. Multilevel modelling of built environment characteristics related to neighbourhood walking activity in older adults. *Journal of Epidemiology & Community Health*, *59*, 558-564.

Li, J., Tian, G., Wu, Z., Zhao, Q., Meng, Y., Xia, Q. 2024. Urban sustainable development: Unraveling regional land development intensity dynamics and future trends in China. *Sustainable Development*, *n/a*, doi:https://doi.org/10.1002/sd.2917.

Li, Z., Ma, J. 2022. Discussing street tree planning based on pedestrian volume using machine learning and computer vision. *Building and Environment*, *219*, 109178, doi:https://doi.org/10.1016/j.buildenv.2022.109178.

Liu, J., Ettema, D., Helbich, M. 2023a. Street view environments are associated with the walking duration of pedestrians: The case of Amsterdam, the Netherlands. *Landscape and Urban Planning*, *235*, 104752, doi:https://doi.org/10.1016/j.landurbplan.2023.104752.

Liu, J., Wang, B., Xiao, L. 2021. Non-linear associations between built environment and active travel for working and shopping: An extreme gradient boosting approach. *Journal of Transport Geography*, *92*, 103034, doi:https://doi.org/10.1016/j.jtrangeo.2021.103034.

Liu, Y., Li, Y., Yang, W., Hu, J. 2023b. Exploring nonlinear effects of built environment on jogging behavior using random forest. *Applied Geography*, *156*, 102990, doi:https://doi.org/10.1016/j.apgeog.2023.102990.

Lu, Y., Sarkar, C., Xiao, Y. 2018. The effect of street-level greenery on walking behavior: Evidence from Hong Kong. *Social Science & Medicine*, *208*, 41-49.

Lundberg, S.M., Lee, S.-I. 2017. A unified approach to interpreting model predictions. *Advances in neural information processing systems*, *30*.

Luo, Z., Qi, X., Sun, C., Dong, Q., Gu, J., Gao, X. 2024. Investigation of influential variations among variables in daylighting glare metrics using machine learning and SHAP. *Building and Environment*, *254*, 111394, doi:https://doi.org/10.1016/j.buildenv.2024.111394.

Ma, X., Chau, C.K., Lai, J.H.K. 2021. Critical factors influencing the comfort evaluation for recreational walking in urban street environments. *Cities*, *116*, 103286, doi:https://doi.org/10.1016/j.cities.2021.103286.

Malambo, P., Kengne, A.P., Lambert, E.V., De Villers, A., Puoane, T. 2017. Association between perceived built environmental attributes and physical activity among adults in South Africa. *BMC public health*, *17*, 1-16.

Marshall Julian, D., Brauer, M., Frank Lawrence, D. 2009. Healthy Neighborhoods: Walkability and Air Pollution. *Environmental Health Perspectives*, *117*, 1752-1759, doi:10.1289/ehp.0900595.

Nagata, S., Nakaya, T., Hanibuchi, T., Amagasa, S., Kikuchi, H., Inoue, S. 2020. Objective scoring of streetscape walkability related to leisure walking: Statistical modeling approach with semantic segmentation of Google Street View images. *Health & Place*, *66*, 102428.

Panter, J., Heinen, E., Mackett, R., Ogilvie, D. 2016. Impact of New Transport Infrastructure on Walking, Cycling, and Physical Activity. *American Journal of Preventive Medicine*, *50*, e45-e53, doi:https://doi.org/10.1016/j.amepre.2015.09.021.

Patel, D., Patel, S., Patel, M., Application of image-to-image translation in improving pedestrian detection. Springer, pp. 471-482.

Porter, A.K., Kohl Iii, H.W., Perez, A., Reininger, B., Gabriel, K.P., Salvo, D. 2018. Peer reviewed: Perceived social and built environment correlates of transportation and recreation-only bicycling among adults. *Preventing chronic disease*, *15*.

Rissel, C., Curac, N., Greenaway, M., Bauman, A., 2012. Physical Activity Associated with Public Transport Use—A Review and Modelling of Potential Benefits, International Journal of Environmental Research and Public Health, pp. 2454-2478.

Salazar-Miranda, A., Zhang, F., Sun, M., Leoni, P., Duarte, F., Ratti, C. 2023. Smart curbs: Measuring street activities in real-time using computer vision. *Landscape and Urban Planning*, *234*, 104715, doi:https://doi.org/10.1016/j.landurbplan.2023.104715.

Shapley, L.S. 1953. A value for n-person games.

Soto, G.W., Webber, B.J., Fletcher, K., Chen, T.J., Garber, M.D., Smith, A., Wilt, G., Conn, M., Whitfield, G.P. 2023. Association between passively collected walking and bicycling data and purposefully collected active commuting survey data—United States, 2019. *Health & place*, *81*, 103002.

Tabatabaie, S., Litt, J.S., Muller, B.H.F. 2023. Sidewalks, trees and shade matter: A visual landscape assessment approach to understanding people's preferences for walking. *Urban Forestry & Urban Greening*, *84*, 127931.

Tainio, M., de Nazelle, A.J., Götschi, T., Kahlmeier, S., Rojas-Rueda, D., Nieuwenhuijsen, M.J., de Sá, T.H., Kelly, P., Woodcock, J. 2016. Can air pollution negate the health benefits of cycling and walking? *Preventive Medicine*, *87*, 233-236, doi:https://doi.org/10.1016/j.ypmed.2016.02.002.

Tainio, M., Jovanovic Andersen, Z., Nieuwenhuijsen, M.J., Hu, L., de Nazelle, A., An, R., Garcia, L.M.T., Goenka, S., Zapata-Diomedi, B., Bull, F., Sá, T.H.d. 2021. Air pollution, physical activity and health: A mapping review of the evidence. *Environment International*, *147*, 105954, doi:https://doi.org/10.1016/j.envint.2020.105954.

Vogel, T., Brechat, P.H., Leprêtre, P.M., Kaltenbach, G., Berthel, M., Lonsdorfer, J. 2009. Health benefits of physical activity in older patients: a review. *International journal of clinical practice*, *63*, 303-320.

Wang, L., Zhao, C., Liu, X., Chen, X., Li, C., Wang, T., Wu, J., Zhang, Y. 2021. Non-linear effects of the built environment and social environment on bus use among older adults in china: An application of the xgboost model. *International Journal of Environmental Research and Public Health*, *18*, 9592.

Wang, R., Lu, Y., Zhang, J., Liu, P., Yao, Y., Liu, Y. 2019. The relationship between visual enclosure for neighbourhood street walkability and elders' mental health in China: Using street view images. *Journal of Transport & Health*, *13*, 90-102, doi:https://doi.org/10.1016/j.jth.2019.02.009.

Wang, X., Liu, Y., Zhu, C., Yao, Y., Helbich, M. 2022. Associations between the streetscape built environment and walking to school among primary schoolchildren in Beijing, China. *Journal of Transport Geography*, *99*, 103303, doi:https://doi.org/10.1016/j.jtrangeo.2022.103303.

Wu, F., Li, W., Qiu, W. 2023. Examining non-linear relationship between streetscape features and propensity of walking to school in Hong Kong using machine learning techniques. *Journal of Transport Geography*, *113*, 103698, doi:https://doi.org/10.1016/j.jtrangeo.2023.103698.

Wu, J., Zhao, C., Li, C., Wang, T., Wang, L., Zhang, Y. 2021. Non-linear Relationships Between the Built Environment and Walking Frequency Among Older Adults in Zhongshan, China. *Frontiers in Public Health*, *9*, doi:10.3389/fpubh.2021.686144.

Xia, C., Yeh, A.G.-O., Zhang, A. 2020. Analyzing spatial relationships between urban land use intensity and urban vitality at street block level: A case study of five Chinese megacities. *Landscape and Urban Planning*, *193*, 103669.

Xiao, L., Lo, S., Liu, J., Zhou, J., Li, Q. 2021. Nonlinear and synergistic effects of TOD on urban vibrancy: Applying local explanations for gradient boosting decision tree. *Sustainable cities and society*, *72*, 103063.

Yang, H., Zhang, Q., Helbich, M., Lu, Y., He, D., Ettema, D., Chen, L. 2022. Examining non-linear associations between built environments around workplace and adults' walking behaviour in Shanghai, China. *Transportation research part A: policy and practice*, *155*, 234-246.

Yang, J., Fricker, P., Jung, A. 2024a. From intangible to tangible: The role of big data and machine learning in walkability studies. *Computers, Environment and Urban Systems*, *109*, 102087, doi:https://doi.org/10.1016/j.compenvurbsys.2024.102087.

Yang, L., Ao, Y., Ke, J., Lu, Y., Liang, Y. 2021a. To walk or not to walk? Examining non-linear effects of streetscape greenery on walking propensity of older adults. *Journal of Transport Geography*, *94*, 103099, doi:https://doi.org/10.1016/j.jtrangeo.2021.103099.

Yang, L., Liu, J., Liang, Y., Lu, Y., Yang, H., 2021b. Spatially Varying Effects of Street Greenery on Walking Time of Older Adults, ISPRS International Journal of Geo-Information.

Yang, W., Fei, J., Li, Y., Chen, H., Liu, Y. 2024b. Unraveling nonlinear and interaction effects of multilevel built environment features on outdoor jogging with explainable machine learning. *Cities*, *147*, 104813, doi:https://doi.org/10.1016/j.cities.2024.104813.

Yang, W., Hu, J., Liu, Y., Guo, W. 2023. Examining the influence of neighborhood and street-level built environment on fitness jogging in Chengdu, China: a massive GPS trajectory data analysis. *Journal of transport geography*, *108*, 103575.

Yasin, S., Iqbal, T., Arshad, Z., Rustam, M., Zafar, M. 2012. Environmental pollution from automobile vehicle service stations. *Journal of quality and technology management*, *8*, 61-70.

Ye, Y., Li, D., Liu, X. 2018. How block density and typology affect urban vitality: An exploratory analysis in Shenzhen, China. *Urban Geography*, *39*, 631-652.

Yin, L., Wang, Z. 2016. Measuring visual enclosure for street walkability: Using machine learning algorithms and Google Street View imagery. *Applied Geography*, *76*, 147-153, doi:https://doi.org/10.1016/j.apgeog.2016.09.024.

Yin, L., Zhang, H. 2021. Building walkable and safe neighborhoods: Assessing the built environment characteristics for pedestrian safety in buffalo, NY. *Journal of Transport & Health*, *22*, 101129, doi:https://doi.org/10.1016/j.jth.2021.101129.

Zafri, N.M., Khan, A. 2022. A spatial regression modeling framework for examining relationships between the built environment and pedestrian crash occurrences at macroscopic level: A study in a developing country context. *Geography and Sustainability*, *3*, 312-324, doi:https://doi.org/10.1016/j.geosus.2022.09.005.

Zhao, P., Li, S., Li, P., Liu, J., Long, K. 2018. How does air pollution influence cycling behaviour? Evidence from Beijing. *Transportation Research Part D: Transport and Environment*, *63*, 826-838, doi:https://doi.org/10.1016/j.trd.2018.07.015.

Zhao, P., Wan, J. 2020. Examining the effects of neighbourhood design on walking in growing megacity. *Transportation Research Part D: Transport and Environment*, *86*, 102417, doi:https://doi.org/10.1016/j.trd.2020.102417.

Zheng, G., Zhang, Y., Yue, X., Li, K. 2023. Interpretable prediction of thermal sensation for elderly people based on data sampling, machine learning and SHapley Additive exPlanations (SHAP). *Building and Environment*, *242*, 110602, doi:https://doi.org/10.1016/j.buildenv.2023.110602.

Zhou, W., Liang, Z., Fan, Z., Li, Z. 2024. Spatio–temporal effects of built environment on running activity based on a random forest approach in nanjing, China. *Health & Place*, *85*, 103176, doi:https://doi.org/10.1016/j.healthplace.2024.103176.